



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 1

Introduction to Forensics

Learning Objectives

- Summarize the basic principles of computer forensics.
- Summarize important laws regarding computer forensics.

Key Concepts

- The chain of custody
- Hardware and networking knowledge used in forensics
- Laws related to computer forensics

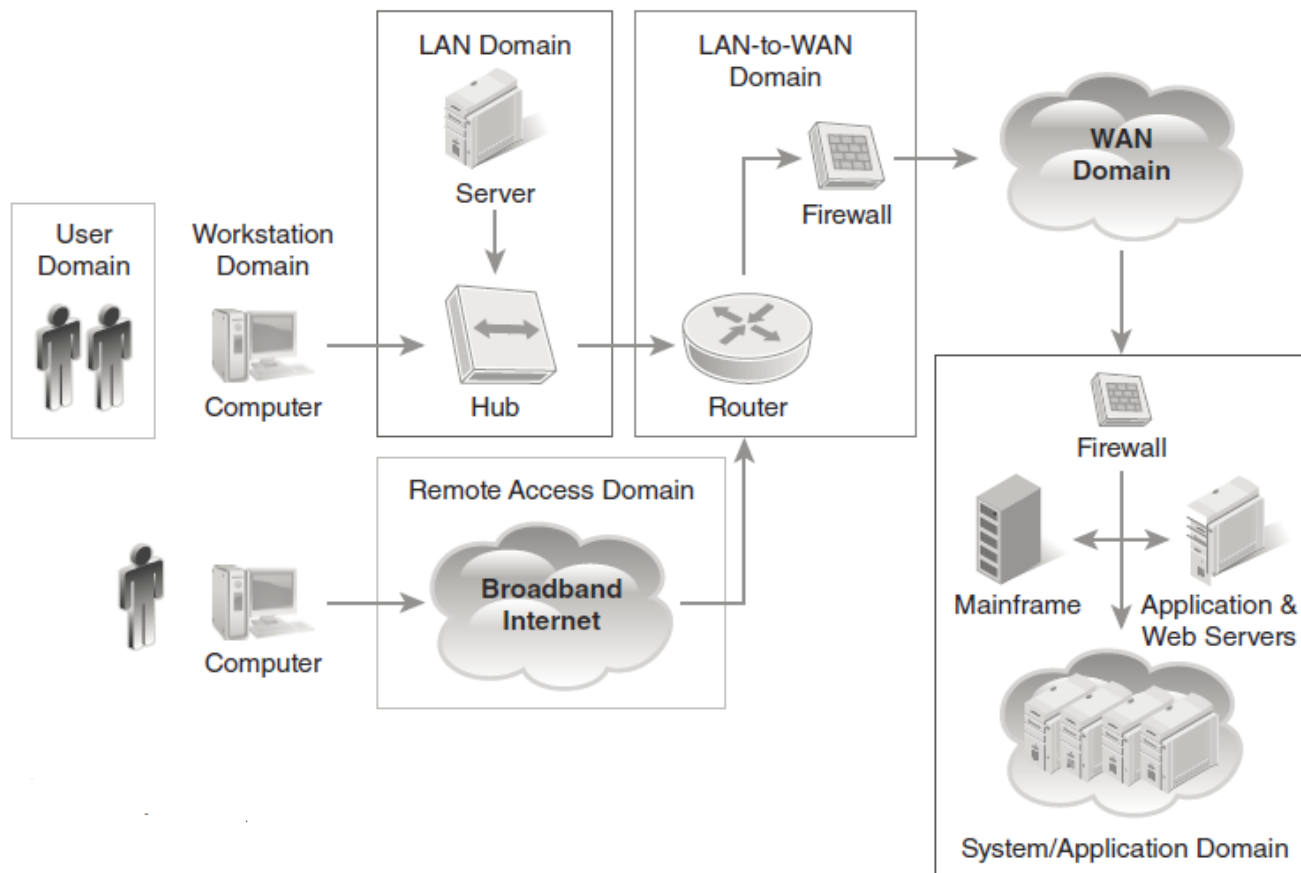
What Is Computer Forensics?

- A subset of forensics
 - *Forensics*: Using “science and technology to investigate and establish facts in criminal or civil courts of law”
- Focuses on extracting data from electronic devices
- Objective is to recover, analyze, and present computer-based material to be used as evidence

What Is Computer Forensics?

- *Digital forensics* is a more encompassing term
 - Includes forensics of cell phones, routers, global positioning system (GPS) devices, tablets, and many other devices
- Can consider computer forensics and digital forensics as interchangeable terms
- Computer forensics applies to all domains of a typical IT infrastructure

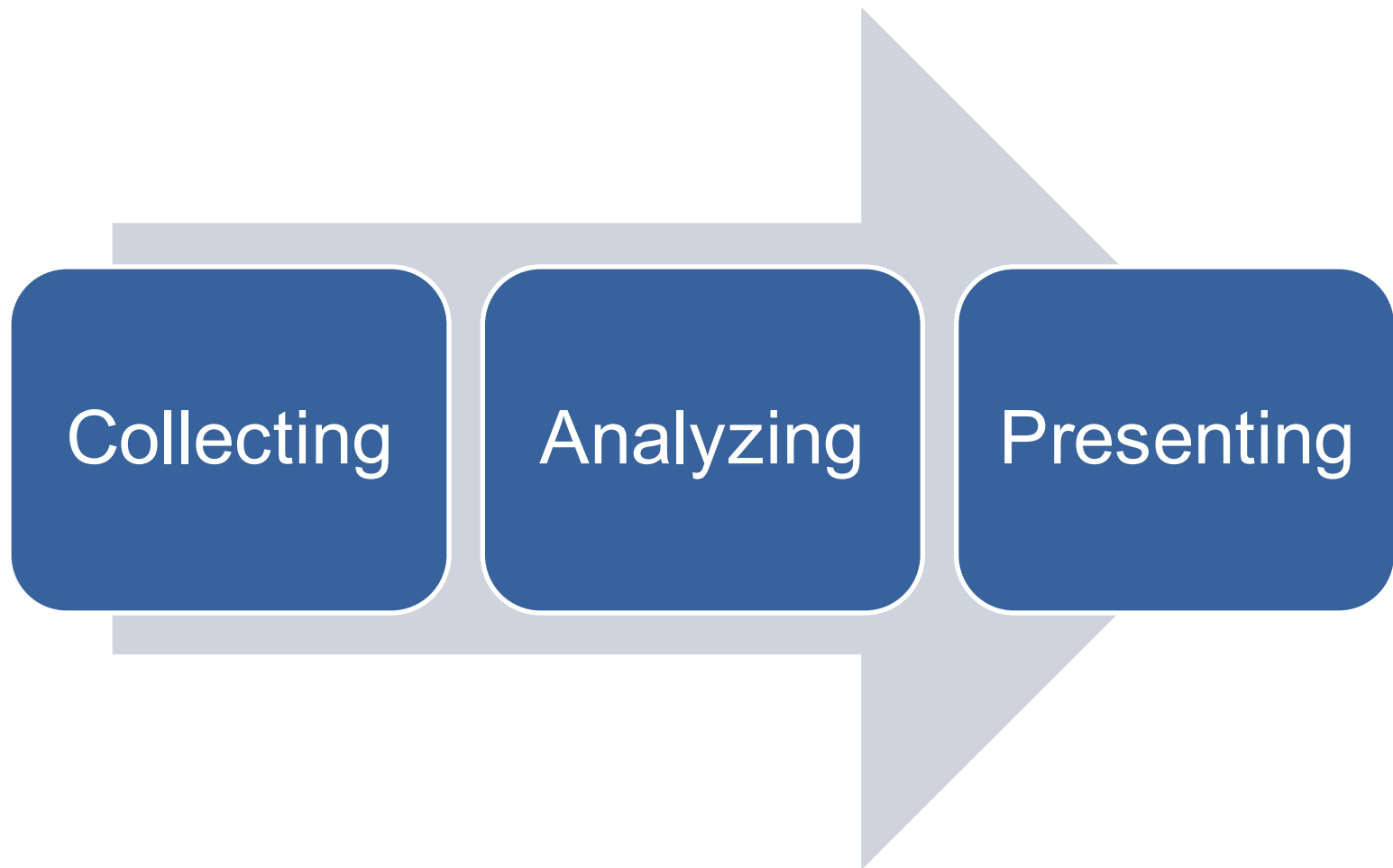
Seven Domains of a Typical IT Infrastructure



Using Scientific Knowledge

- Computer forensics is a science
- You apply scientific methods and processes
- You must have:
 - Knowledge of relevant scientific disciplines
 - Scientific knowledge of the field
- Computer forensics begins with a thorough understanding of computer hardware

Process in Computer Forensics



Collecting

- Maintain the chain of custody from point of seizure to the moment it is shown in court by recording where it was, how it was stored, and who had access to it.
- Don't touch the suspect drive. Make a forensic copy and work with the copy.
- Create a document trail.
- Secure the evidence.

Analyzing

- Look at every detail.
- Ask what each detail means.
- Make sure you have evidence to support each conclusion.
- Consider possible alternative conclusions.

Presenting

- Write an expert report that lists what tests you conducted, what you found, and your conclusions.
- Testify as an expert witness in a disposition or in the actual trial.

The Field of Digital Forensics

- Involves processing and assembling data so that it becomes relevant to an investigation
- Data that supports a specific finding or determination is *evidence*

Who Uses Computer Forensics?

The military

Government
agencies

Law firms

Criminal
prosecutors

Academia

Data
recovery
firms

Corporations

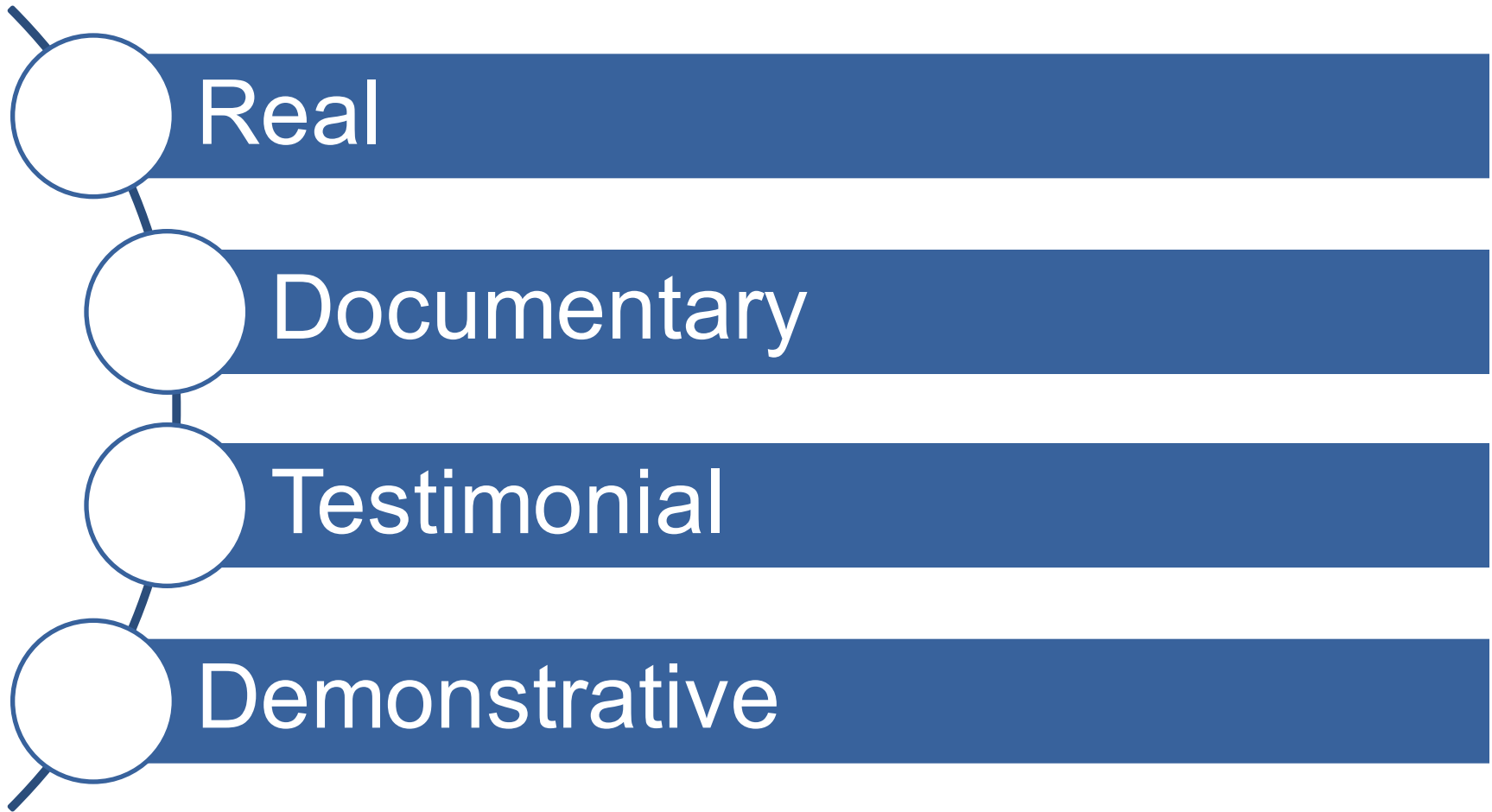
Insurance
companies

Individuals

What Is Digital Evidence?

- Information that has been processed and assembled so that it is relevant to an investigation and supports a specific finding or determination
- Raw information is not, in and of itself, evidence
- Data has to be relevant to a case in order to be evidence

Types of Evidence



Scope-Related Challenges

Large volumes of data

System complexity

Distributed crime scenes

Growing caseloads/limited resources

Types of Digital System Forensics Analysis

Physical
storage
media

Email

Network

Internet

Software

Live system

Cell phone/
mobile

General Guidelines

Maintain chain of custody

Don't touch suspect drive

Create a document trail

Secure evidence

Knowledge Needed for Forensic Analysis

Hardware

Software

Networks

Addresses

Computer Hardware

- Memory
- Hard drives/storage devices



Software: Operating Systems

- Windows
- Linux
- Macintosh

Files and File Systems

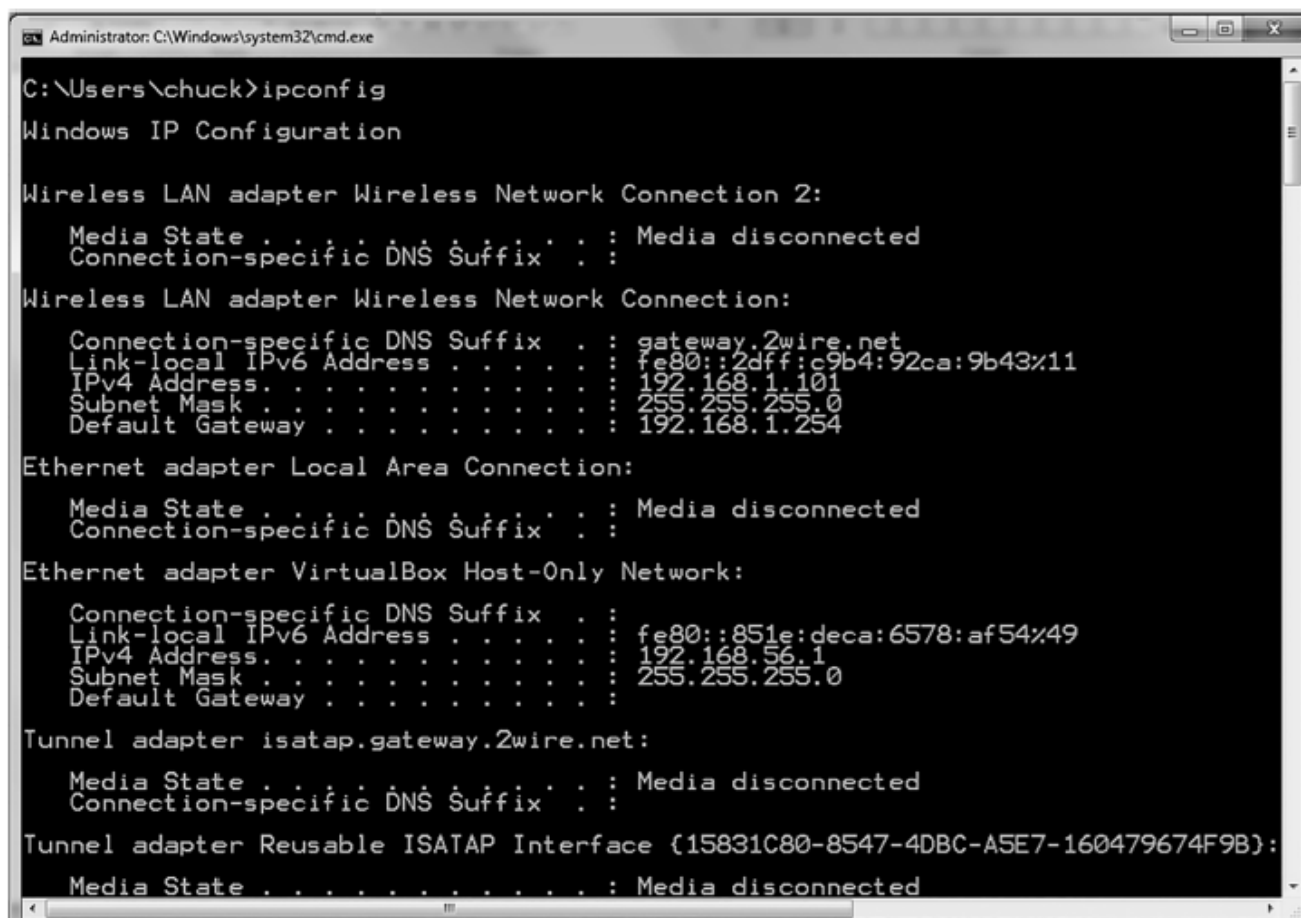
- File
 - Any document, spreadsheet, picture, video, or even program
- File systems
 - Physical journaling
 - Logical journaling



Networking: Addressing

- Physical ports
- MAC address
- IP address
 - IPv4 example: 192.2.132.8
- Port number
- Uniform resource locator (URL)

Networking Utilities: ipconfig



```
Administrator: C:\Windows\system32\cmd.exe

C:\Users\chuck>ipconfig

Windows IP Configuration

Wireless LAN adapter Wireless Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
Wireless LAN adapter Wireless Network Connection:

    Connection-specific DNS Suffix  . : gateway.2wire.net
    Link-local IPv6 Address . . . . . : fe80::2dff:c9b4:92ca:9b43%11
    IPv4 Address. . . . . : 192.168.1.101
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.254

Ethernet adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
Ethernet adapter VirtualBox Host-Only Network:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::851e:deca:6578:af54%49
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

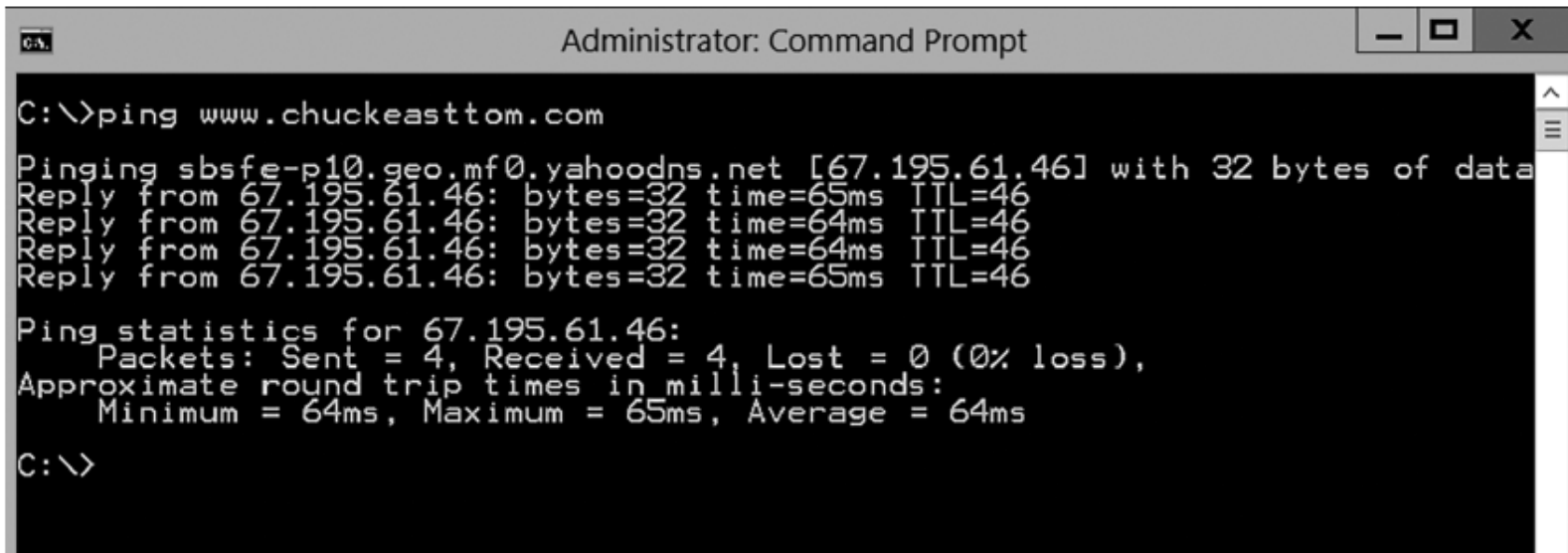
Tunnel adapter isatap.gateway.2wire.net:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
Tunnel adapter Reusable ISATAP Interface {15831C80-8547-4DBC-A5E7-160479674F9B}:

    Media State . . . . . : Media disconnected
```

Used with permission from Microsoft.

Networking Utilities: ping



```
C:\>ping www.chuckeasttom.com

Pinging sbsfe-p10.geo.mf0.yahoodns.net [67.195.61.46] with 32 bytes of data:
Reply from 67.195.61.46: bytes=32 time=65ms TTL=46
Reply from 67.195.61.46: bytes=32 time=64ms TTL=46
Reply from 67.195.61.46: bytes=32 time=64ms TTL=46
Reply from 67.195.61.46: bytes=32 time=65ms TTL=46

Ping statistics for 67.195.61.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 64ms, Maximum = 65ms, Average = 64ms

C:\>
```

Used with permission from Microsoft.

Networking Utilities: tracert

Used with permission from Microsoft.

```
Administrator: Command Prompt

C:\>tracert www.chuckeasttom.com

Tracing route to sbsfe-p10.geo.mf0.yahoodns.net [67.195.61.46]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    Tardis [192.168.1.1]
  2  <1 ms    <1 ms    <1 ms    192.168.0.1
  3  8 ms     8 ms     7 ms     142.254.141.53
  4  26 ms    20 ms    28 ms    tge0-0-4.plaotxso01h.texas.rr.com [24.28.90
  5  12 ms    9 ms     10 ms    agg21.plantxmp01r.texas.rr.com [24.175.49.2
  6  13 ms    13 ms    14 ms    agg27.crtntxjt01r.texas.rr.com [24.175.36.1
  7  46 ms    13 ms    13 ms    agg21.dllatx1301r.texas.rr.com [24.175.49.0
  8  11 ms    15 ms    14 ms    66.109.1.216
  9  10 ms    9 ms     10 ms    107.14.17.133
 10  10 ms    12 ms    11 ms    UNKNOWN-216-115-102-X.yahoo.com [216.115.10
 11  54 ms    64 ms    54 ms    ae-3.pat2.bfz.yahoo.com [216.115.97.209]
 12  64 ms    64 ms    64 ms    ae-6.pat2.gqb.yahoo.com [216.115.96.62]
 13  66 ms    64 ms    64 ms    et-18-1-0.msr2.gq1.yahoo.com [66.196.67.115
 14  64 ms    64 ms    65 ms    et-1-0-0.clr2-a-gdc.gq1.yahoo.com [67.195.3
 15  64 ms    64 ms    64 ms    te-8-1.bas2-1-flk.gq1.yahoo.com [67.195.1.1
 16  64 ms    65 ms    64 ms    p10pn-i.geo.vip.gq1.yahoo.com [67.195.61.46

Trace complete.

C:\>_
```

Obscured Information and Anti-Forensics

- Obscured information
 - May be scrambled by encryption, hidden using steganographic software, compressed, or a proprietary format
 - Cybercriminals sometimes obscure information to deter forensic examination
 - Companies use certain manipulation and storage techniques to protect business-sensitive information

Obscured Information and Anti-Forensics (Cont.)

■ Anti-forensics

- Attackers may use techniques to intentionally conceal their identities, locations, and behavior
- Examples:
 - Perpetrators may conceal their identities by using networked connections at a library
 - Perpetrators may use encryption or anonymous services to protect themselves

The Daubert Standard

- Legal precedent set in the 1993 Supreme Court case of *Daubert v. Merrill Dow Pharmaceuticals* [509 U.S. 579 (1993)]
- Regards the admissibility of expert witnesses' testimony during legal proceedings
- Any scientific evidence presented in a trial has to have been reviewed and tested by the relevant scientific community

Daubert Standard (Cont.)

Empirical testing

Peer review

Error rate and standards

Generally accepted theory/technique

Laws Important to Digital Forensics



- Federal Privacy Act of 1974
- Privacy Protection Act of 1980 (PPA)
- Communications Assistance for Law Enforcement Act of 1994
- Electronic Communications Privacy Act of 1986
- Computer Security Act of 1987

Laws Important to Digital Forensics (Cont.)



- Foreign Intelligence Surveillance Act of 1978
- Child Protection and Sexual Predator Punishment Act of 1998
- Children's Online Privacy Protection Act of 1988
- Communications Decency Act of 1996

Laws Important to Digital Forensics (Cont.)



- Telecommunications Act of 1996
- Wireless Communications and Public Safety Act of 1999
- USA Patriot Act
- Sarbanes-Oxley Act of 2002

Laws Important to Digital Forensics (Cont.)



- 18 U.S.C. § 1030: Fraud and Related Activity in Connection with Computers
- 18 U.S.C. § 1020: Fraud and Related Activity in Connection with Access Devices
- The Digital Millennium Copyright Act (DMCA) of 1998
- 18 U.S.C. § 1028A: Identity Theft and Aggravated Identity Theft
- 18 U.S.C. § 2251: Sexual Exploitation of Children

The Communications Assistance for Law Enforcement Act (CALEA) of 1994

- A federal wiretap law for traditional wired telephony
- Was expanded to include wireless, Voice over Internet Protocol (VoIP), and other forms of electronic communications
 - Including signaling traffic and metadata

The Foreign Intelligence Surveillance Act of 1978

- A law that allows for the collection of “foreign intelligence information” between foreign powers and agents of foreign powers using physical and electronic surveillance
- FISA court issues a warrant for actions under FISA

The Child Protection and Sexual Predator Punishment Act of 1998

- Requires service providers that become aware of the storage or transmission of child pornography to report it to law enforcement

The Wireless Communications and Public Safety Act of 1999

- Allows for the collection and use of “empty” communications
 - Nonverbal and nontext communications, such as GPS information

The USA Patriot Act of 2001

- The primary law under which a wide variety of Internet and communications information content and metadata is currently collected
- Provisions exist within the Patriot Act to protect identity and privacy of U.S. citizens

Warrants/Search and Seizure/Fourth Amendment

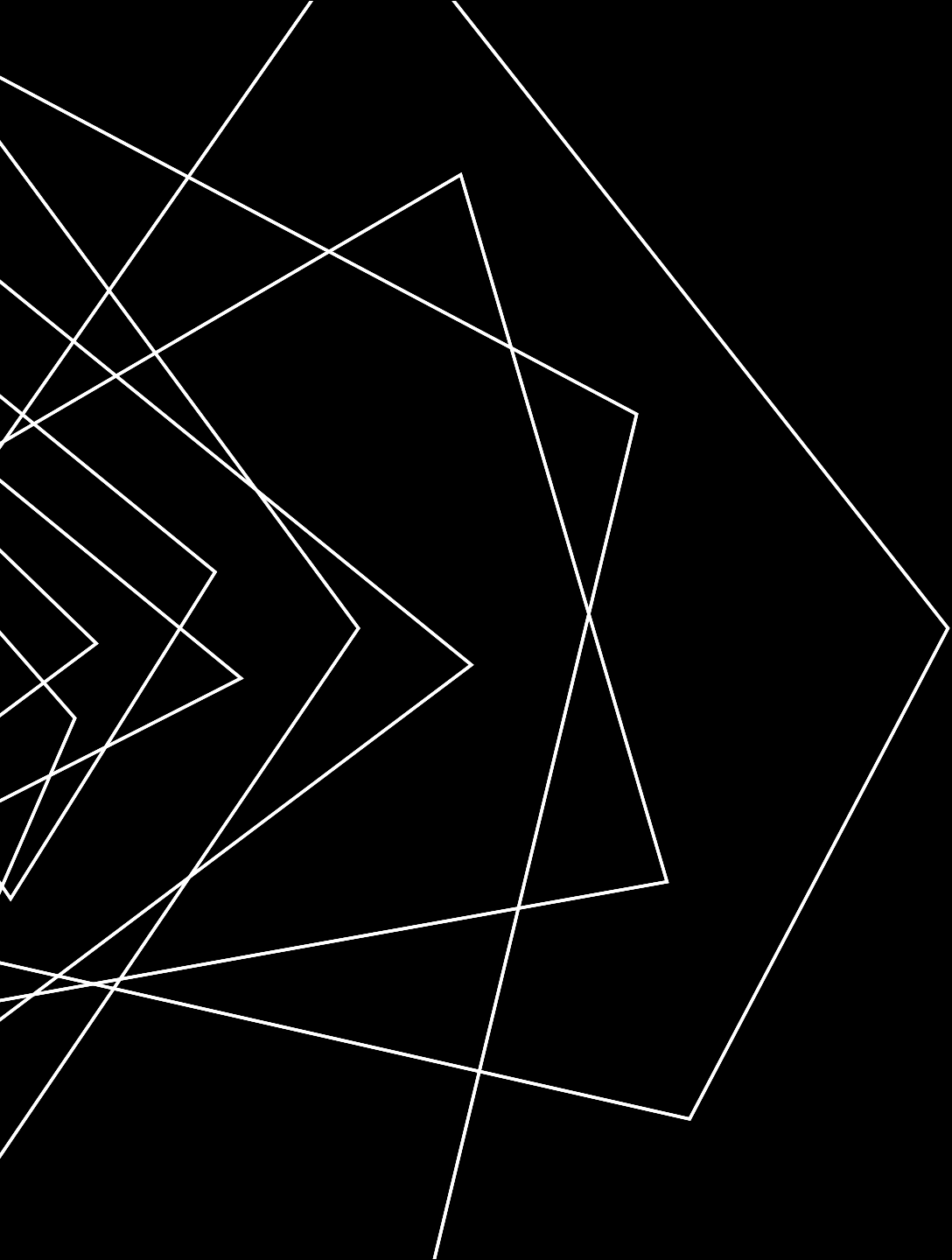
- Warrants
- Search and seizure
- Fourth Amendment and reasonable expectation of privacy

Federal Guidelines

- FBI computer forensics guidelines
- Secret Service "golden rules"
- Regional Computer Forensics Laboratory (RCFL) Program

Summary

- Basic concepts of computer forensics, including the chain of custody
- The process of computer forensics
- Types of evidence
- Hardware, software, networking, and address knowledge used in forensics
- Laws related to computer forensics



THANK YOU

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